CSA6303- Threat Intelligence and Network Security

AT1: Application-Based Questions

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats |Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational) — Threat Intelligence Lifecycle — Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

Dr.V.Parthipan **Total: 50**

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Code of Conduct:CSA6303

I C Amareswar Reddy /192412470 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : C Amareswar Reddy

AT1: Application-Based Questions

Q1.CIA Triad — Scenario Identification (5 marks)
An e-commerce site stays fully online during an incident, but customers notice product prices have been silently changed by an attacker who gained database write access. No downtime occurred. Which CIA Triad principle(s) are violated? Justify your answer

Q2. Risk Quantification — Annualized Loss Expectancy (10 marks)
A company's customer database is valued at $50,000. A breach would destroy/corrupt 40% of its value (Exposure Factor). Historical data shows this type of breach occurs about twice a year (Annualized Rate of Occurrence = 2). Calculate the Single Loss Expectancy (SLE) and the Annualized Loss Expectancy (ALE). Show all formulas used.

Q3. Brute-Force Attack Time Estimation (10 marks)
An 8-character password uses a 62-character set (a-z, A-Z, 0-9). An attacker's cracking rig tests 1 billion passwords per second. Calculate the size of the keyspace and the worst-case time required to exhaust it. Express your answer in hours.

Q4. Risk Prioritization Matrix (10 marks)
Given five threats rated on Likelihood (1-5) and Impact (1-5), compute the Risk Score (Likelihood x Impact) for each and rank them in order of patching/budget priority:
(a) Ransomware — L4, I5;
(b) Phishing — L5, I3;
(c) Insider data theft — L2, I5;
(d) DDoS — L3, I3;
(e) Unpatched server exploit — L4, I4.

Q5. Passive vs. Active Threat Classification (5 marks)
Classify each of the following as a passive or active threat, and state which CIA principle it primarily threatens:
(a) An attacker silently sniffing unencrypted traffic on a public Wi-Fi network.
(b) An attacker performing ARP spoofing to redirect traffic.
(c) A nation-state actor performing long-term network reconnaissance without altering anything.

Q6. Threat Intelligence Lifecycle sequencing (10 marks)
A SOC analyst receives an executive request: "Are we at risk from the ransomware group currently targeting our industry?" Arrange the following into the correct order of the Threat Intelligence Lifecycle, and name each phase:
(i) Analysts correlate collected IOCs with internal logs.
(ii) A report is sent to the CISO.
(iii) The request itself is logged as a requirement.
(iv) OSINT and threat feeds are pulled for the group's known TTPs. (v) The CISO's follow-up questions shape the next collection cycle.

Q2. Risk Quantification — Annualized Loss Expectancy (ALE) (10 Marks)

Given Data:

- Asset Value (AV) = $50,000
- Exposure Factor (EF) = 40% = 0.4
- Annualized Rate of Occurrence (ARO) = 2 times per year

Step 1: Calculate Single Loss Expectancy (SLE)

Definition: Single Loss Expectancy (SLE) is the amount of financial loss expected from a single security incident.

Formula:

Substitute the given values:

Therefore, the expected loss from a single breach is **$20,000**.

Step 2: Calculate Annualized Loss Expectancy (ALE)

Definition: Annualized Loss Expectancy (ALE) is the total expected financial loss due to a risk over one year.

Formula:

Substitute the values:

Therefore, the expected annual financial loss due to this type of breach is **$40,000**.

Final Answer

Parameter	**Formula**	**Result**
Asset Value (AV)	Given	$50,000
Exposure Factor (EF)	Given	0.4
Annualized Rate of Occurrence (ARO)	Given	2
Single Loss Expectancy (SLE)	AV × EF	$20,000
Annualized Loss Expectancy (ALE)	SLE × ARO	$40,000

Conclusion: The company can expect a loss of **$20,000 per breach** and an annualized loss of **$40,000** if the breach occurs twice a year.

Q3. Brute-Force Attack Time Estimation (10 Marks)

Given Data:

- Password length = 8 characters
- Character set = 62 characters (26 lowercase letters + 26 uppercase letters + 10 digits)
- Password testing speed = 1 billion passwords per second

We need to calculate:

1. The total password keyspace.
2. The worst-case time required to test every possible password, expressed in hours.

Step 1: Calculate the Size of the Keyspace

Definition: Keyspace is the total number of possible password combinations that an attacker must test to find the correct password.

Formula:

Where:

- = Number of possible characters = 62
- = Password length = 8

Substituting the values:

Therefore, the total number of possible passwords is:

Step 2: Calculate the Worst-Case Cracking Time in Seconds

Definition: Worst-case cracking time is the time required to test all possible password combinations before finding the correct password in the final attempt.

Formula:

Substituting the values:

Step 3: Convert Seconds into Hours

We know that:

Formula:

Substituting the value:

Final Answer

Parameter	**Value**
Password length	8 characters
Character set size	62 characters
Total keyspace	218,340,105,584,896
Testing speed	1 billion passwords/second
Worst-case time in seconds	218,340.1056 seconds
Worst-case time in hours	**60.65 hours**

Conclusion: An attacker testing passwords per second would require approximately **60.65 hours (about 2.53 days)** to exhaust the entire keyspace of an 8-character password using 62 possible characters, assuming a constant testing speed.